



Department of Computer Science

National University of Computer and Emerging Sciences
Lahore Campus

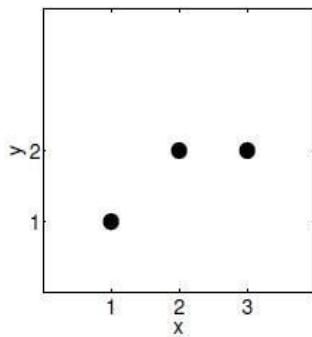
Data Mining

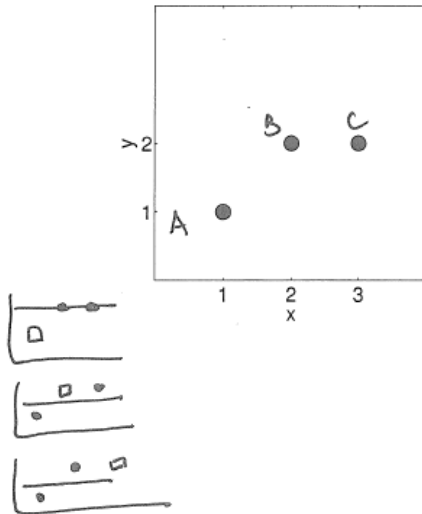
Section A

Quiz 1

Question 1: 4 marks

1) Compute the leave-one-out cross-validation error of a zero-order predictor.





$$\begin{aligned}
 A: & \text{predict } 2 \Rightarrow (1-2)^2 = 1^2 \\
 B: & \text{predict } 1\frac{1}{2} \Rightarrow (2-1\frac{1}{2})^2 \Rightarrow + (\frac{1}{2})^2 \\
 C: & \text{predict } 1\frac{1}{2} \Rightarrow \quad \quad \quad + (\frac{1}{2})^2
 \end{aligned}$$

$$\Rightarrow \text{MSE} = \frac{1}{3}(1 + \frac{1}{4} + \frac{1}{4}) = \frac{1}{3}(1\frac{1}{2}) = \frac{1}{2}$$

Question 2: 3 marks

Suppose that we have training data $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})\}$ and we wish to predict y using a nonlinear regression model with two parameters:

$$\hat{y} = a + \log(b x_1)$$

We decide to train our model using gradient descent on the mean squared error (MSE).

Write down the gradient of the MSE.

(a) Write down the expression for the MSE on our training set.

$$J(b) = \frac{1}{m} \sum_i (y^i - \hat{y}^i)^2 = \frac{1}{m} \sum_i (y^i - (a + \log(b x^i)))^2$$

(b) Write down the gradient of the MSE.

$$\nabla J = \left[\frac{\partial J}{\partial a} \quad \frac{\partial J}{\partial b} \right]$$

$$\hat{y}^i = a + \log b x^i$$

$$\frac{\partial J}{\partial a} = \frac{1}{m} \sum_i (2) [y^i - \hat{y}^i] (-1)$$

$$\frac{\partial J}{\partial b} = \frac{1}{m} \sum_i (2) [y^i - \hat{y}^i] (-1) \frac{1}{\log b x^i} \cdot x^i$$

Question 3: 3 marks

Which of the following are closed item sets? Also, write down your reasoning for why you chose these item sets as closed items.

		Items									
		A	B	C	D	E	F	G	H	I	J
Transactions	1										
	2										
	3										
	4										
	5										
	6										
	7										
	8										
	9										
	10										

Closed itemsets: {C,D,E,F}, {C,F}

